

1. Let $\mathcal{C}$ be the curve described by the vector function $\vec{r}: [0, \frac{3\pi}{4}]$ defined by

$$\vec{r}(t) = (3 \sin(2t), 2 \cos(2t)) \quad \text{with} \quad 0 \leq t \leq \frac{3\pi}{4}.$$

- (i) Sketch the curve $\mathcal{C}$ indicating the points of intersection with the axes.
 - (ii) Find the parametric equations of the tangent line to $\mathcal{C}$ at the point $(\frac{3\sqrt{2}}{2}, -\sqrt{2})$.
2. Let $\mathcal{C}$ be the curve described by the vector function $\mathbf{r}(t) = (2 \sin(t\sqrt{2}), t, 2 \cos(t\sqrt{2}))$ with $0 \leq t \leq \frac{\pi}{\sqrt{2}}$. Show that the length of the arc of the curve $\mathcal{C}$ from $(2, \frac{\pi}{2\sqrt{2}}, 0)$ to $(0, \frac{\pi}{\sqrt{2}}, -2)$ is $L = \frac{3\pi}{2\sqrt{2}}$.
3. Given two real numbers a and b both different from 0.

- (i) Find the arc length parameterization $\vec{\gamma}(s)$ of the curve described by $\vec{r}(t) = (a \cos t, a \sin t, bt)$ with $0 \leq t \leq 2\pi$.
- (ii) Show that $\vec{\gamma}'(s)$ is a unit vector.
- (iii) Find the scalar product $\vec{\gamma}'(s) \cdot \vec{\gamma}''(s)$.
- (iv) Where you expecting this value? Explain clearly your answer. (Hint: $\vec{\gamma}'(s) \cdot \vec{\gamma}'(s) = 1$ and differentiate both sides of this equation.)

4. Consider the curve $\mathcal{C}$ described by the vector function $\vec{r}: [0, 2\pi] \rightarrow \mathbb{R}^3$ defined by

$$\vec{r}(t) = (e^t \sin t, e^{2t} \cos t, 2), \quad t \in [0, 2\pi].$$

- (a) Find $\vec{r}'(\pi/2)$.
- (b) Find the parametric equations of the tangent line to $\mathcal{C}$ at the point $(e^{\pi/2}, 0, 2)$.
- (c) At which point(s) is the tangent line parallel to the y -axis?

5. Given the curve $\mathcal{C}$ described by the vector function $\vec{r}: [0, 2\pi] \rightarrow \mathbb{R}^3$ defined by

$$\vec{r}(t) = (2 \sin(2t), t, 2 \cos(2t)) \quad t \in [0, 2\pi].$$

Find:

- (i) the unit tangent vector, the curvature and the principal unit normal vector to the curve $\mathcal{C}$ at the point $(\sqrt{3}, \pi/6, 1)$.

- (ii) The binormal vector $\vec{B}(\frac{\pi}{6})$.
 - (v) the cartesian equations of the normal plane and the osculating plane to the curve $\mathcal{C}$ at the point $(\sqrt{3}, \pi/6, 1)$.
6. Show that the arc length parametrization of the curve described by the vector function $\vec{r}(t) := (e^t \cos t, e^t \sin t, e^t)$, $0 \leq t \leq \pi$ is given by

$$\vec{\gamma}(s) = (1 + s/\sqrt{3})(\cos(\ln(1 + s/\sqrt{3})), \sin(\ln(1 + s/\sqrt{3})), 1), \quad 0 \leq s \leq \sqrt{3}(e^\pi - 1).$$